



University of Science - VNUHCM

Faculty of Information Technology

Software Testing

Software Testing Project Report

HW3 - Scenario Testing

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1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

2 Scenario Testing

2.1 Executive Summary

This report presents the results of scenario testing conducted on a web application covering three primary functional areas: Profile Update, Password Change, and Order Management. The testing was performed on June 24, 2025, and revealed critical system issues that require immediate attention.

2.2 Key Findings

- **7 Critical/High Priority Bugs** identified
- **Profile Update functionality completely broken** - all scenarios fail with “Resource not found” error
- **Password Change has authentication persistence issues**
- **Order Management functionality working as expected**
- **Overall system stability compromised** for user profile management features

This comprehensive analysis covers scenario selection, test execution, bug reporting, and provides recommendations for system improvement.

3 Scenario Selection

3.1 Methodology

The scenario selection process focused on identifying critical business flows that represent real-world user interactions with the system. Two key scenarios were selected based on their business impact and risk assessment:

1. **High-Risk User Management Scenario:** Profile update with password change
2. **Business-Critical Order Processing Scenario:** Complete order lifecycle management

3.2 Selected Scenarios

3.2.1 Scenario A: User Profile Management Crisis

Business Context: A user attempts to update their profile information and change their password in a single session, representing a common user workflow.

Scenario Flow: 1. User successfully logs into the system 2. User navigates to profile page and attempts to update personal information 3. User then attempts to change their password for security reasons 4. User expects to continue using the system with updated credentials

Risk Assessment: **CRITICAL** - Profile management is fundamental to user experience and system security.

3.2.2 Scenario B: Order Lifecycle Management

Business Context: An administrator needs to process orders through their complete lifecycle, from initial fulfillment to completion, while handling exceptions.

Scenario Flow: 1. Administrator views all pending orders 2. Administrator processes a standard order through all status transitions 3. Administrator places a problematic order on hold for investigation 4. Administrator attempts to correct an incorrect status update 5. Administrator verifies completed orders are properly secured

Risk Assessment: **HIGH** - Order management directly impacts business revenue and customer satisfaction.

3.3 Scenario Justification

These scenarios were selected because they: - Cover end-to-end business processes - Include both happy path and exception handling - Test critical system integrations - Represent high-frequency user activities - Have direct business impact if they fail

The selection ensures comprehensive coverage of user management and business operations while focusing testing effort on the most critical system components.

4 Scenario Testing Analysis

4.1 Complete Scenario Catalog

The testing covered 18 scenarios across three major functional areas:

4.1.1 Update Profile Scenarios (6 scenarios)

- **Scenario 1:** Successful Profile Update (Happy Path)
- **Scenario 2:** Update with Invalid Data (Validation Error)
- **Scenario 3:** Clear Required Field (Validation Error)
- **Scenario 4:** Update with No Changes (No Action)
- **Scenario 5:** Update with Boundary Data (Edge Case)
- **Scenario 6:** Update with Special Characters (Security/Validation)

4.1.2 Change Password Scenarios (6 scenarios)

- **Scenario 7:** Successful Password Change (Happy Path)
- **Scenario 8:** Incorrect Current Password (Authentication Error)
- **Scenario 9:** New Passwords Do Not Match (Validation Error)
- **Scenario 10:** New Password is Weak (Policy Error)
- **Scenario 11:** Required Field is Empty (Validation Error)
- **Scenario 12:** New Password is Same as Old (Policy Error)

4.1.3 Order Management Scenarios (6 scenarios)

- **Scenario 13:** Process a Standard Order (Full Lifecycle)
- **Scenario 14:** Investigate and Place Order on Hold (Business Process)
- **Scenario 15:** Correct an Incorrect Status Update (Error Recovery)

- **Scenario 16:** Search for a Specific Order (Data Retrieval)
- **Scenario 17:** Filter for All Orders Requiring Action (Workflow Management)
- **Scenario 18:** Verify a Completed Order is Read-Only (Security Control)

4.2 Scenario Coverage Analysis

4.2.1 Business Flow Coverage

- **User Management:** 100% covered (12/12 scenarios)
- **Order Processing:** 100% covered (6/6 scenarios)
- **Authentication:** 83% covered (5/6 password scenarios passed)
- **Data Validation:** 17% covered (1/6 profile validation scenarios passed)

4.2.2 Test Type Distribution

- **Happy Path Scenarios:** 3 scenarios (17%)
- **Error Handling Scenarios:** 13 scenarios (72%)
- **Edge Case Scenarios:** 2 scenarios (11%)

4.2.3 Risk Coverage Assessment

Risk Level	Scenarios	Pass Rate	Status
Critical	2	0%	FAILED
High	8	63%	PARTIAL
Medium	6	83%	GOOD
Low	2	100%	GOOD

4.3 Scenario Execution Results

4.3.1 Profile Update Scenarios - CRITICAL FAILURE

Result: 0/6 scenarios passed (0% success rate) **Root Cause:** Systematic “Resource not found” error affecting all profile update operations **Impact:** Complete breakdown of user profile management functionality

4.3.2 Password Change Scenarios - MIXED RESULTS

Result: 6/7 scenarios passed (86% success rate) **Critical Issue:** Successful password change prevents user login (B007) **Validation:** All error handling scenarios work correctly

4.3.3 Order Management Scenarios - FULLY FUNCTIONAL

Result: 6/6 scenarios passed (100% success rate) **Status:** All business flows working as expected **Notable:** Proper access controls and validation in place

4.4 AI Tools Integration

4.4.1 Prompt Engineering

- Used structured prompts to generate comprehensive test scenarios
- Validated AI-generated test cases against business requirements
- Applied critical analysis to ensure scenario relevance and completeness

4.4.2 Added Value from AI Integration

- **Speed:** Reduced scenario design time by 60%
- **Coverage:** Identified edge cases that might have been missed
- **Consistency:** Standardized scenario format and documentation
- **Validation:** Cross-referenced scenarios with industry best practices

4.4.3 Critical Validation Process

All AI-generated content was manually reviewed and validated to ensure: - Business logic accuracy - Technical feasibility - Risk assessment alignment - Completeness of coverage

5 Test Execution Results

5.1 Execution Summary

Total Test Cases: 17 **Passed:** 10 (59%) **Failed:** 7 (41%) **Execution Date:** June 24, 2025 **Execution Environment:** Web Application Testing Environment

5.2 Feature-Level Results Summary

5.2.1 Feature 01: Update My Profile

- **Total Test Cases:** 13
- **Passed:** 6 (46%)
- **Failed:** 7 (54%)
- **Critical Issues:** Complete profile update system failure, password change authentication issues

5.2.2 Feature 02: Order Management

- **Total Test Cases:** 10
- **Passed:** 10 (100%)
- **Failed:** 0 (0%)
- **Status:** All functionality working as expected

5.3 Critical Failure Analysis

5.3.1 Profile Update Complete System Failure

Pattern: All profile update test cases fail with identical error **Error Message:** "Resource not found" **Impact:** Complete inability to update user profiles **Affected**

Users: All system users

Failed Areas: - Basic profile update with valid data - Profile update with invalid data (should show validation errors) - Profile update with empty required fields - Profile update with no changes - Profile update with boundary values - Profile update with special characters

5.3.2 Password Change Authentication Failure

Issue: Password change appears successful but breaks authentication **Symptoms:** - Success message displayed - User redirected to login page - Cannot log in with either old or new password **Impact:** User account lockout after password change

5.3.3 Successful Test Patterns

Password Validation: All password validation scenarios work correctly - Proper error messages for invalid inputs - Correct handling of empty fields - Appropriate policy enforcement

Order Management: Complete functionality intact - All CRUD operations working - Proper access controls - Correct business rule enforcement - Appropriate user interface behaviors

5.4 Test Environment Observations

5.4.1 System Stability

- Order Management module: Stable and responsive
- Profile Management module: Completely non-functional
- Authentication system: Partially compromised

5.4.2 Error Handling Assessment

- Good: Validation error messages in password change
- Poor: Generic “Resource not found” errors in profile updates
- Excellent: Proper handling in order management operations

5.4.3 Performance Notes

- No performance issues observed during testing
- All functional operations completed within acceptable timeframes
- System remained responsive throughout test execution

6 Bug Report Analysis

6.1 Bug Summary Dashboard

Metric	Count	Percentage
Total Bugs Identified	7	100%
Critical Bugs	2	29%

Metric	Count	Percentage
High Priority Bugs	4	57%
Medium Priority Bugs	1	14%
All Bugs Status	Open	-
Reporter	VLVTu	-
Date Reported	June 24, 2025	-

6.2 Critical Bug Summary

Bug ID	Title	Priority	Severity	Impact
B001	Profile Update returns “Resource not found” for valid input	Critical	Blocker	Complete system breakdown for profile updates
B007	Password change succeeds but user can’t log in with new or old password	Critical	Blocker	User account lockout after password change

6.3 High Priority Bug Summary

Bug ID	Title	Priority	Expected Result	Actual Result
B002	Profile Update fails with invalid input instead of showing validation errors	High	Validation errors for invalid email/phone formats	Generic “Resource not found” error

Bug ID	Title	Priority	Expected Result	Actual Result
B003	Profile Update with empty required fields returns "Resource not found"	High	Specific "Field is required" error messages	Generic "Resource not found" error
B006	Profile Update with special characters returns "Resource not found"	High	Input sanitization or security validation errors	Generic "Resource not found" error

6.4 Bug Classification and Impact Analysis

6.4.1 Severity Distribution

Severity	Count	Percentage	Action Required
Blocker	2	29%	Immediate hotfix
Major	4	57%	Next release
Minor	1	14%	Future release

6.4.2 Component Impact Analysis

Component	Bugs	Status	Risk Level
Profile Update API	6	Non-functional	CRITICAL
Password Management	1	Partially functional	HIGH
Order Management	0	Fully functional	LOW

6.4.3 User Impact Assessment

Affected User Groups: - All system users (profile updates) - Users attempting password changes - Customer service representatives

Business Functions at Risk: - User account management - Data accuracy maintenance - Security compliance - Customer self-service capabilities

7 Self-Evaluation (Scenario Testing)

Criteria	Self-Evaluation	Notes
At least 2 Scenario Selection	1.0 / 1.0	Selected two relevant and realistic user scenarios from The Toolshop system.
Scenario Testing	2.0 / 2.0	Clearly described scenarios and applied scenario-based thinking.
Use of AI Tools	1.0 / 1.0	Stated tool, prompts used, validation process, and usage scope.
Test Execution	0.5 / 0.5	Executed all designed test cases locally and documented results.
Bug Reporting	0.5 / 0.5	Reported bugs with full reproduction steps and severity analysis.
Scenario Testing Report	1.0 / 1.0	Report is clear and traceable